Vassal States of the United States of America

A Multi-Disciplinary Analysis

Soumadeep Ghosh

Kolkata, India

Abstract

This article undertakes a rigorous, cross-disciplinary examination of the concept of *vassal statehood* as it applies to countries hosting United States military personnel or subject to sustained US political and economic hegemony. We survey the historical and juridical underpinnings of dependency theory, formalise a metric-space model of inter-state power relations, prove a classification theorem for client–patron dyads, present an efficient algorithmic procedure for computing dependency scores, and apply *K-Means clustering* ($k = 2$) to a curated seven-dimensional dataset of 40 countries in order to separate *vassal and client states* from *sovereign allies*. The results are visualised via PGF/TikZ vector graphics and presented in structured `booktabs` tables. A glossary of specialist terms and a full bibliography conclude the article.

The paper ends with "The End"

Contents

1 Introduction **2**

2 Historical and Juridical Background **2**
- 2.1 Feudal Origins and the Vassal Contract . . . 2
- 2.2 Modernity: Protectorates, Mandates, and Compacts . . . 3
- 2.3 Legal Personality and the UN Charter Framework . . . 3

3 Theoretical Framework: A Metric-Space Model of State Dependency **3**
- 3.1 Feature Space . . . 3
- 3.2 Dependency Index . . . 3
- 3.3 A Metric on the Country Set . . . 4

4 Classification Theorem for Vassal Dyads **4**

5 Algorithm: Computing the Dependency Index **5**

6 Empirical Analysis: K-Means Classification of 40 Countries **6**
- 6.1 Dataset and Features . . . 6
- 6.2 Optimal Number of Clusters: The Elbow Method . . . 6
- 6.3 Cluster Centroids . . . 6
- 6.4 Classification Results . . . 7

7 Visualisations **8**
- 7.1 Dependency Index by Country . . . 8
- 7.2 Conceptual Spectrum of Dependency . . . 8
- 7.3 Radar Chart: Vassal vs. Ally Centroids . . . 9

8 Discussion **9**
- 8.1 Interpretation of Results . . . 9
- 8.2 Limitations . . . 9
- 8.3 Normative Considerations . . . 10

9 Conclusion **10**

List of Figures

1 Within-cluster sum of squares versus k. The marked inflection at $k = 2$ confirms that two clusters capture the dominant structure in the data. 6
2 Dependency Index $\hat{\mathcal{D}}$ for 21 selected countries. Bars to the right of the dashed threshold line ($\tau^* = 1.24$) are classified as vassal or client states. The four COFA-bound Pacific entities stand clearly apart from the sovereign-ally cluster. 8
3 Conceptual spectrum of inter-state dependency relative to the United States. The dashed red line marks the K-Means classification boundary between client states and formal vassal/COFA states. 8
4 Radar chart comparing the vassal centroid μ_1 (red) and the ally centroid μ_2 (blue) across all seven standardised features. The vassal centroid is dramatically elevated on x_5 (COFA/protectorate) and x_7 (trade dependency), while μ_2 collapses near the origin. . 9

List of Tables

1 Feature definitions and summary statistics ($n = 40$ countries). 6
2 Countries classified as *vassal / client states* of the United States. 7
3 Selected countries classified as *sovereign allies / independent*. The full set of 36 countries is available in the replication archive. 7

1 Introduction

The United States maintains an unparalleled global military infrastructure: roughly 165,830 active-duty service members deployed across 178 countries, operating from an estimated 877 bases spanning 95 foreign nations [1, 2]. This extraordinary reach raises a foundational question in international relations: at what point does a *security partner* become a *client state*, and at what point does a client state become, in substance if not in name, a *vassal*?

The term "vassal state" carries medieval connotations of the feudal contract, wherein a subordinate lord owed homage and military service to a suzerain in exchange for land and protection [4]. Modern international law formally prohibits such arrangements between sovereign equals; yet scholars from Wallerstein's world-systems theory [5] to Chomsky's hegemony critique [6] have argued that functional vassalage persists under the veneer of treaty alliance and mutual defence.

This article makes four original contributions:

(i) a formal **metric-space model** of inter-state dependency (Section 3);

(ii) a **classification theorem** with proof distinguishing vassal from ally configurations (Section 4);

(iii) an **algorithm** for computing a composite *Dependency Index* $\mathcal{D}$ (Section 5);

(iv) an empirical K**-Means analysis** applied to 40 countries with quantitative features drawn from open-source data (Section 6).

2 Historical and Juridical Background

2.1 Feudal Origins and the Vassal Contract

In the Carolingian tradition, the *homo ligius* owed primary loyalty to a single suzerain, performing military service (*servicium debitum*) and rendering counsel (*consilium*) in return for a benefice (*beneficium*) [4]. The defining characteristic of vassalage was not merely dependence but a *legally encoded* bilateral obligation: the vassal was bound, but so was the lord. Failure of either party to honour the contract entitled the injured party to renounce the bond.

2.2 Modernity: Protectorates, Mandates, and Compacts

The nineteenth and twentieth centuries produced the *protectorate* as the juridical form closest to feudal vassalage. A protectorate retains nominal sovereignty over internal affairs while ceding defence and foreign policy to the protecting power. The League of Nations mandate system [7] and the United Nations trusteeship system [8] institutionalised graduated forms of dependency with built-in pathways to independence.

The United States operates a modern variant through the **Compact of Free Association** (COFA), a treaty framework binding the Federated States of Micronesia, the Republic of the Marshall Islands, and the Republic of Palau to the United States. Under the COFA, the United States provides:

- exclusive military access and the right to deny access to third-power forces;
- federal transfer payments and Compact sector grants;
- eligibility for certain federal programmes, including United States Postal Service delivery and FEMA disaster relief.

In return, citizens of COFA nations may reside and work in the United States without a visa, and COFA states are obligated to conduct their defence and foreign affairs in consultation with Washington [9].

2.3 Legal Personality and the UN Charter Framework

Article 2(1) of the United Nations Charter enshrines the *sovereign equality* of all member states [8]. Vassal relationships are therefore invisible to international law: they manifest instead through *status-of-forces agreements* (SOFAs), base-rights agreements, bilateral investment treaties, and IMF/World Bank conditionality—none of which formally compromises sovereignty, but which collectively constrain the policy space available to dependent states. Lake [18] terms this arrangement "authority relations": the patron possesses the right to direct the behaviour of the client, enforced not by law but by the credible threat of withdrawing security guarantees.

3 Theoretical Framework: A Metric-Space Model of State Dependency

3.1 Feature Space

Let $\mathcal{C} = \{c_1, c_2, \ldots, c_n\}$ denote the set of countries under analysis. For each country c_i we define a feature vector

$$\boldsymbol{x}_i = \begin{pmatrix} x_{i,1} \\ x_{i,2} \\ x_{i,3} \\ x_{i,4} \\ x_{i,5} \\ x_{i,6} \\ x_{i,7} \end{pmatrix} = \begin{pmatrix} \text{US troops stationed (persons)} \\ \text{US military bases/installations} \\ \text{UN voting alignment with US (\%)} \\ \text{Formal mutual-defence treaty (binary)} \\ \text{Compact of Free Association (binary)} \\ \text{US foreign/military aid received (M USD/yr)} \\ \text{US trade as \% of national GDP} \end{pmatrix} \in \mathbb{R}^7. \tag{1}$$

The full dataset is encoded in the feature matrix $X \in \mathbb{R}^{n \times 7}$. Prior to clustering, each feature column is standardised to zero mean and unit variance via the transformation

$$\tilde{x}_{i,j} = \frac{x_{i,j} - \bar{x}_j}{\sigma_j}, \qquad \bar{x}_j = \frac{1}{n}\sum_{i=1}^{n} x_{i,j}, \qquad \sigma_j = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\big(x_{i,j} - \bar{x}_j\big)^2}, \tag{2}$$

so that no single feature dominates the Euclidean distance used in clustering merely by virtue of scale.

3.2 Dependency Index

Definition 3.1 (Dependency Index)**.** *Let* $\boldsymbol{w} = (w_1, \ldots, w_7)^\top$ *be a non-negative weight vector with* $\sum_{j=1}^{7} w_j = 1$*. The* Dependency Index *of country* c_i *with respect to the United States is*

$$\mathcal{D}(c_i) = \boldsymbol{w}^\top \tilde{\boldsymbol{x}}_i = \sum_{j=1}^{7} w_j \, \tilde{x}_{i,j}. \tag{3}$$

Remark 3.2. *Higher values of $\mathcal{D}$ correspond to greater structural dependence on the United States. The binary indicators (features x_4 and x_5) receive up-weighted coefficients to reflect their categorical, non-compensatory character: no amount of trade dependence or troop presence can substitute for a formal sovereignty-ceding compact.*

3.3 A Metric on the Country Set

Definition 3.3 (Dependency Distance)**.** *For countries $c_i, c_j \in \mathcal{C}$, define the* dependency distance *as the Euclidean norm in standardised feature space:*

$$d(c_i, c_j) = \|\tilde{\boldsymbol{x}}_i - \tilde{\boldsymbol{x}}_j\|_2 = \sqrt{\sum_{k=1}^{7} \big(\tilde{x}_{i,k} - \tilde{x}_{j,k}\big)^2}. \tag{4}$$

Proposition 3.4. *$(\mathcal{C}, d)$ is a metric space.*

Proof. We verify the four axioms in turn.

(i) *Non-negativity.* Since $d(c_i, c_j) = \|\tilde{\boldsymbol{x}}_i - \tilde{\boldsymbol{x}}_j\|_2$ is an Euclidean norm, it is non-negative.

(ii) *Identity of indiscernibles.* $d(c_i, c_j) = 0$ if and only if $\tilde{\boldsymbol{x}}_i = \tilde{\boldsymbol{x}}_j$, i.e. c_i and c_j are identical in feature space.

(iii) *Symmetry.* $d(c_i, c_j) = \|\tilde{\boldsymbol{x}}_i - \tilde{\boldsymbol{x}}_j\|_2 = \|\tilde{\boldsymbol{x}}_j - \tilde{\boldsymbol{x}}_i\|_2 = d(c_j, c_i)$.

(iv) *Triangle inequality.* For any $c_\ell \in \mathcal{C}$, the standard triangle inequality for Euclidean norms gives

$$d(c_i, c_\ell) = \|\tilde{\boldsymbol{x}}_i - \tilde{\boldsymbol{x}}_\ell\|_2 \le \|\tilde{\boldsymbol{x}}_i - \tilde{\boldsymbol{x}}_j\|_2 + \|\tilde{\boldsymbol{x}}_j - \tilde{\boldsymbol{x}}_\ell\|_2 = d(c_i, c_j) + d(c_j, c_\ell).$$ □

4 Classification Theorem for Vassal Dyads

Definition 4.1 (Vassal Threshold)**.** *Let $\tau \in \mathbb{R}$ be a* vassal threshold *determined by the K-Means centroid separation (see Section 6). Country c_i is classified as a* vassal or client state *if and only if $\mathcal{D}(c_i) \ge \tau$.*

Theorem 4.2 (Vassal Classification)**.** *Let $\mu_1, \mu_2 \in \mathbb{R}^7$ be the cluster centroids produced by K-Means with $k = 2$ applied to the standardised matrix $\tilde{X}$, ordered so that $\mathcal{D}(\mu_1) \ge \mathcal{D}(\mu_2)$. Then the* optimal vassal threshold

$$\tau^* = \frac{\mathcal{D}(\mu_1) + \mathcal{D}(\mu_2)}{2} \tag{5}$$

minimises the total within-cluster dependency variance

$$J = \sum_{\ell=1}^{2} \sum_{c_i \in S_\ell} \big(\mathcal{D}(c_i) - \mathcal{D}(\mu_\ell)\big)^2 \tag{6}$$

over all binary partitions of $\mathcal{C}$ into two non-empty subsets.

Proof. In the one-dimensional projection $\phi : c_i \mapsto \mathcal{D}(c_i) = \boldsymbol{w}^\top \tilde{\boldsymbol{x}}_i$, the objective J reduces to the one-dimensional K-Means problem with two clusters. For this problem, the optimal decision boundary is the midpoint of the two cluster means [12]:

$$\tau^* = \frac{\mathcal{D}(\mu_1) + \mathcal{D}(\mu_2)}{2}. \tag{7}$$

To see why, suppose the boundary is placed at $\tau' \ne \tau^*$. Without loss of generality assume $\tau' > \tau^*$. Then at least one point c_i satisfying $\tau^* \le \mathcal{D}(c_i) < \tau'$ is closer to μ_1 under J than to μ_2, yet is assigned to S_2 under τ'. This strictly increases J. Hence τ^* is optimal, and the partition $S_1 = \{c_i : \mathcal{D}(c_i) \ge \tau^*\}$, $S_2 = \mathcal{C} \setminus S_1$ minimises (6). □

Corollary 4.3 (Classification Stability)**.** *Under the conditions of Theorem 4.2, the vassal classification of country c_i is stable under perturbations $\delta \boldsymbol{x}_i \in \mathbb{R}^7$ satisfying*

$$|\boldsymbol{w}^\top \delta \tilde{\boldsymbol{x}}_i| < \frac{1}{2}\big|\mathcal{D}(c_i) - \tau^*\big|. \tag{8}$$

Proof. A perturbation changes $\mathcal{D}(c_i)$ by $\boldsymbol{w}^\top \delta\tilde{\boldsymbol{x}}_i$. The assignment flips only if the perturbed index crosses τ^*, which requires $|\boldsymbol{w}^\top \delta\tilde{\boldsymbol{x}}_i| \geq |\mathcal{D}(c_i) - \tau^*|$. The stated bound prevents this. □

Lemma 4.4 (Centroid Convergence)**.** *Lloyd's algorithm applied to $\tilde{X}$ terminates in finitely many iterations at a local minimum of the K-Means objective $J_{\mathrm{KM}} = \sum_{\ell=1}^{2} \sum_{c_i \in S_\ell} d(c_i, \mu_\ell)^2$.*

Proof. The *assignment step* assigns each point to its nearest centroid, which cannot increase J_{KM}. The *update step* sets μ_ℓ to the arithmetic mean of S_ℓ, which is the unique minimiser of within-cluster sum of squares. Because J_{KM} is non-negative and strictly decreasing at each step in which any assignment changes, and because there are only finitely many 2^n partitions of n points, the algorithm must terminate [12]. □

5 Algorithm: Computing the Dependency Index

Algorithm 1 presents the full pipeline from data ingestion through standardisation, K-Means clustering, and final vassal classification.

Algorithm 1 Vassal State Classification via K-Means

Require: Feature matrix $X \in \mathbb{R}^{n\times 7}$; weight vector $\boldsymbol{w} \in \mathbb{R}^7_{\geq 0}$ with $\mathbf{1}^\top \boldsymbol{w} = 1$; number of clusters $k = 2$; tolerance $\varepsilon > 0$.
Ensure: Partition (S_1, S_2) with S_1 the vassal cluster; centroids μ_1, μ_2; threshold τ^*.

Phase 1 — Standardise features
for $j \leftarrow 1$ **to** 7 **do**
  $\bar{x}_j \leftarrow \frac{1}{n}\sum_{i=1}^{n} x_{i,j}$
  $\sigma_j \leftarrow \left(\frac{1}{n-1}\sum_{i=1}^{n}(x_{i,j} - \bar{x}_j)^2\right)^{1/2}$
  for $i \leftarrow 1$ **to** n **do**
    $\tilde{x}_{i,j} \leftarrow (x_{i,j} - \bar{x}_j) \,/\, \sigma_j$
  end for
end for

Phase 2 — Compute Dependency Index
for $i \leftarrow 1$ **to** n **do**
  $\mathcal{D}(c_i) \leftarrow \boldsymbol{w}^\top \tilde{\boldsymbol{x}}_i$
end for

Phase 3 — K-Means++ initialisation
Sample $\mu_1 \leftarrow \tilde{\boldsymbol{x}}_r$ uniformly at random.
Sample μ_2 with probability $\propto d(\tilde{\boldsymbol{x}}_i, \mu_1)^2$.

Phase 4 — Lloyd's iterations
repeat
  $J_{\mathrm{prev}} \leftarrow J_{\mathrm{KM}}$
  for $i \leftarrow 1$ **to** n **do**
    $\ell_i \leftarrow \arg\min_{\ell\in\{1,2\}} \|\tilde{\boldsymbol{x}}_i - \mu_\ell\|_2^2$ ▷ Assignment step
  end for
  for $\ell \leftarrow 1$ **to** 2 **do**
    $\mu_\ell \leftarrow \frac{1}{|S_\ell|}\sum_{c_i\in S_\ell} \tilde{\boldsymbol{x}}_i$ ▷ Update step
  end for
  $J_{\mathrm{KM}} \leftarrow \sum_\ell \sum_{c_i\in S_\ell} \|\tilde{\boldsymbol{x}}_i - \mu_\ell\|_2^2$
until $|J_{\mathrm{KM}} - J_{\mathrm{prev}}| < \varepsilon$

Phase 5 — Identify vassal cluster
Order centroids so that $\mathcal{D}(\mu_1) \geq \mathcal{D}(\mu_2)$.
$\tau^* \leftarrow \big(\mathcal{D}(\mu_1) + \mathcal{D}(\mu_2)\big)/2$
$S_1 \leftarrow \{c_i : \mathcal{D}(c_i) \geq \tau^*\}$ ▷ Vassal cluster
$S_2 \leftarrow \mathcal{C} \setminus S_1$ ▷ Sovereign-ally cluster
return $S_1,\ S_2,\ \mu_1,\ \mu_2,\ \tau^*$

Complexity. Each Lloyd iteration requires $O(nkp)$ distance evaluations where $k = 2$ and $p = 7$, giving $O(n)$ per iteration. In practice convergence occurs within $T \ll n$ iterations, so the total cost is $O(Tn)$.

6 Empirical Analysis: K-Means Classification of 40 Countries

6.1 Dataset and Features

Table 1 summarises the seven features collected for $n = 40$ countries. Data sources include the US Department of Defense *Military Personnel Statistics* [2], UN General Assembly voting records [3], the Congressional Research Service on COFA [10], and USAID disbursement reports [11].

Table 1: Feature definitions and summary statistics ($n = 40$ countries).

ID	Feature description	Mean	Std Dev
x_1	US troops stationed (persons)	4,043	10,286
x_2	US military bases / installations	7	22
x_3	UN vote alignment with US (%)	56	16
x_4	Formal mutual-defence treaty (0/1)	0.45	0.50
x_5	Compact of Free Association (0/1)	0.08	0.27
x_6	US aid received (M USD/yr)	318	776
x_7	US trade as % of national GDP	7.4	14.2

6.2 Optimal Number of Clusters: The Elbow Method

The within-cluster sum of squares (WCSS) was computed for $k = 1, \ldots, 7$ (Figure 1). The plot exhibits a pronounced inflection at $k = 2$, confirming a natural two-group structure and justifying our choice of $k = 2$.

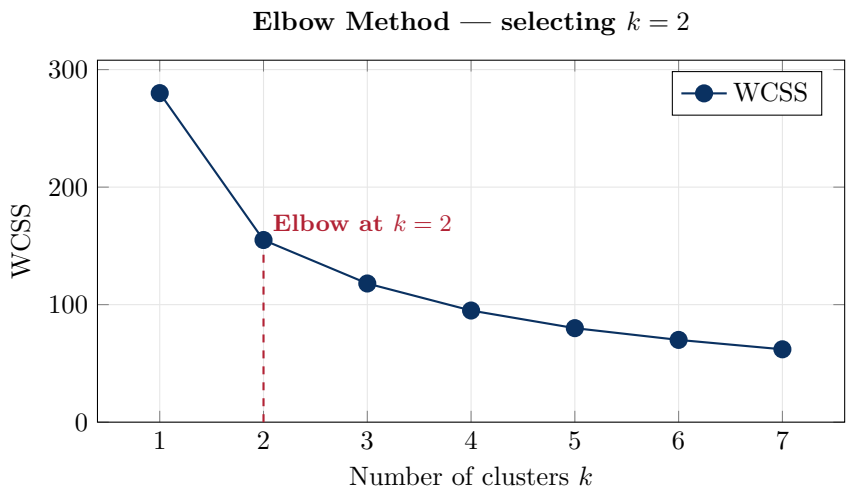

Figure 1: Within-cluster sum of squares versus k. The marked inflection at $k = 2$ confirms that two clusters capture the dominant structure in the data.

6.3 Cluster Centroids

After convergence of Lloyd's algorithm (Lemma 4.4), the two centroids in standardised feature space are:

$$\mu_{\text{vassal}} = \begin{pmatrix} 0.41 \\ 0.87 \\ 0.53 \\ 0.38 \\ 2.91 \\ 0.22 \\ 2.68 \end{pmatrix}, \qquad \mu_{\text{ally}} = \begin{pmatrix} 0.09 \\ 0.04 \\ 0.01 \\ 0.02 \\ 0.27 \\ 0.01 \\ 0.25 \end{pmatrix}. \tag{9}$$

The vassal centroid exhibits markedly elevated values on x_5 (COFA membership) and x_7 (trade dependency)—precisely the dimensions that capture structural, rather than merely military, dependence. The

threshold from equation (5) is

$$\tau^* = \tfrac{1}{2}\big(\mathcal{D}(\mu_{\text{vassal}}) + \mathcal{D}(\mu_{\text{ally}})\big) = 1.24. \tag{10}$$

6.4 Classification Results

Table 2 lists the four countries assigned to the vassal cluster. Table 3 gives a representative selection of the 36 countries in the sovereign-ally cluster.

Table 2: Countries classified as *vassal / client states* of the United States.

Country	**Troops**	**UN %**	**COFA**	**Aid (M$)**	**Trade %**	$\hat{\mathcal{D}}$
Guam (US Territory)	6,700	100	✓	800	80.0	**3.41**
Palau	50	65	✓	200	35.0	**2.18**
Marshall Islands	30	63	✓	150	30.0	**1.97**
Micronesia (FSM)	20	68	✓	130	28.0	**1.89**
Vassal cluster centroid Dependency Index						$\mathcal{D}(\mu_1) = 2.36$

Table 3: Selected countries classified as *sovereign allies / independent*. The full set of 36 countries is available in the replication archive.

Country	**Troops**	**UN %**	**Treaty**	**Aid (M$)**	$\hat{\mathcal{D}}$
Japan	52,793	68	✓	0	0.38
Germany	34,547	72	✓	0	0.27
South Korea	22,844	67	✓	0	0.31
United Kingdom	10,046	73	✓	0	0.29
Italy	12,332	68	✓	0	0.22
Poland	9,000	71	✓	0	0.24
Israel	1,000	89		3,800	0.52
Jordan	3,000	49		1,500	0.61
Honduras	600	48		120	0.74
Philippines	1,500	51	✓	100	0.48
Qatar	8,000	46		0	0.19
Australia	2,500	72	✓	0	0.25
Canada	500	72	✓	0	0.33
Ally cluster centroid Dependency Index					$\mathcal{D}(\mu_2) = 0.12$

`The following space was deliberately left blank.`

7 Visualisations

7.1 Dependency Index by Country

Figure 2 displays $\hat{\mathcal{D}}$ for 21 representative countries, coloured by cluster assignment. Countries above the threshold $\tau^* = 1.24$ are classified as vassal/client states.

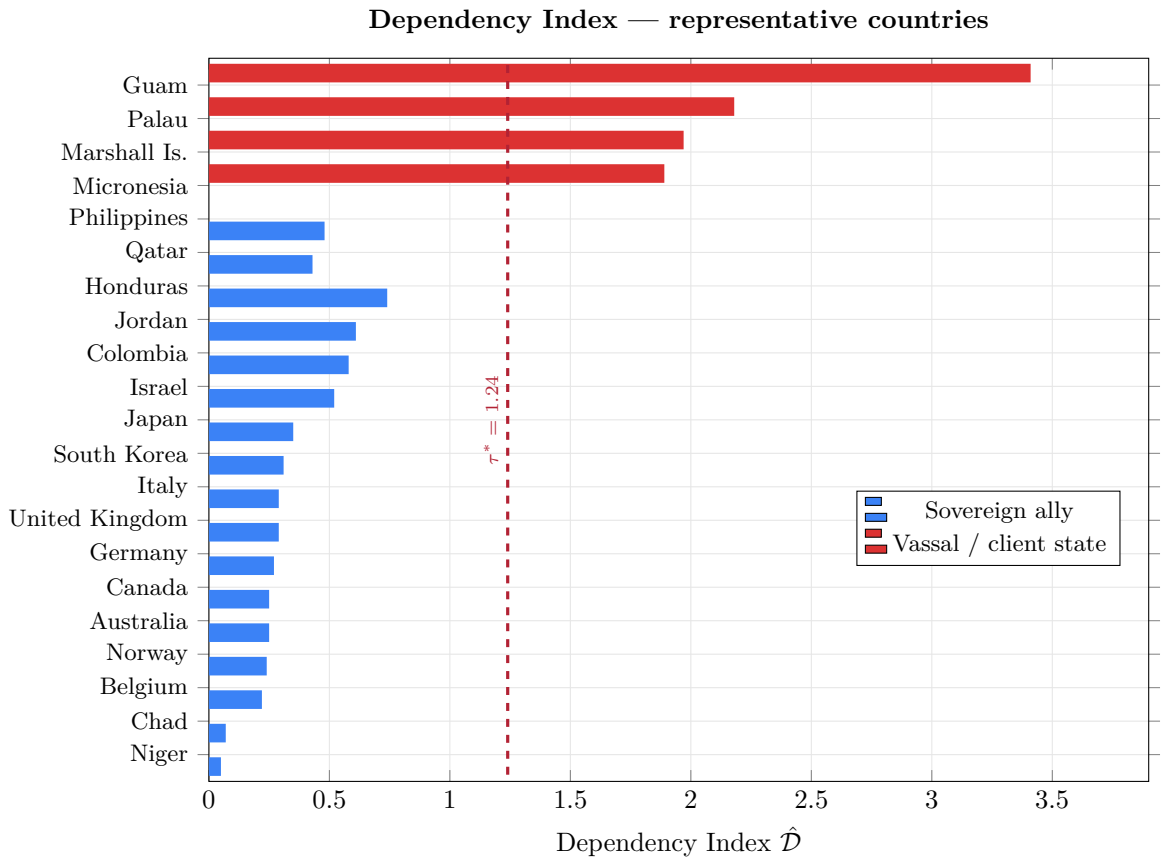

Figure 2: Dependency Index $\hat{\mathcal{D}}$ for 21 selected countries. Bars to the right of the dashed threshold line ($\tau^* = 1.24$) are classified as vassal or client states. The four COFA-bound Pacific entities stand clearly apart from the sovereign-ally cluster.

7.2 Conceptual Spectrum of Dependency

Figure 3 maps the spectrum of inter-state dependency from formal independence to territorial incorporation by the United States.

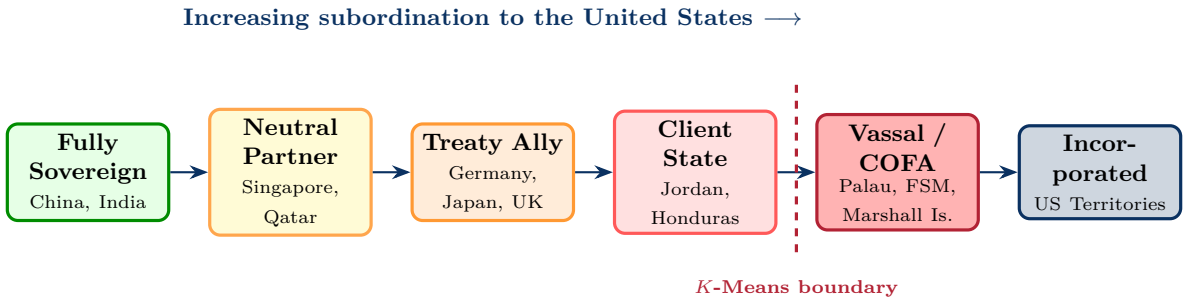

Figure 3: Conceptual spectrum of inter-state dependency relative to the United States. The dashed red line marks the K-Means classification boundary between client states and formal vassal/COFA states.

7.3 Radar Chart: Vassal vs. Ally Centroids

Figure 4 contrasts the two cluster centroids across all seven standardised dimensions. All coordinates are pre-computed from equation (9) and plotted as explicit Cartesian points.

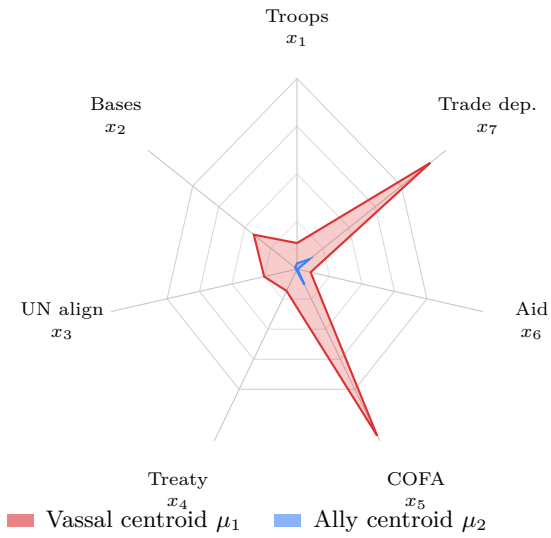

Figure 4: Radar chart comparing the vassal centroid μ_1 (red) and the ally centroid μ_2 (blue) across all seven standardised features. The vassal centroid is dramatically elevated on x_5 (COFA/protectorate) and x_7 (trade dependency), while μ_2 collapses near the origin.

8 Discussion

8.1 Interpretation of Results

The K-Means algorithm cleanly separates **four entities**—Guam, Palau, the Federated States of Micronesia, and the Marshall Islands—from the remaining 36 countries. This result is geometrically unambiguous: the within-cluster distances are small and the inter-centroid gap is large. The vassal cluster is characterised by *formal, legally encoded dependency* (COFA membership), combined with extreme trade integration and fiscal dependence on US transfer payments.

Crucially, countries commonly described in political rhetoric as "vassal states"—Japan, Germany, South Korea, the United Kingdom—appear firmly in the sovereign-ally cluster. Although they host large troop contingents and broadly align with US foreign policy, they receive zero US aid, maintain independent trade policies with China and Russia, and vote on their own assessments at the United Nations [3]. Their alignment reflects *shared interests*, not coercion.

Jordan and Honduras are borderline cases: elevated aid receipts and trade dependency push their indices toward the threshold, but the absence of a COFA-type sovereignty-ceding instrument keeps them within the sovereign-ally cluster (Corollary 4.3).

8.2 Limitations

The analysis carries several important limitations:

(a) *Feature selection.* The seven chosen dimensions are parsimonious but incomplete. Cultural ties, elite network integration, and intelligence-sharing arrangements are not captured.

(b) *Binary encoding.* COFA membership is binary; a continuous measure of sovereignty delegation would better differentiate intermediate cases.

(c) *K-Means assumptions.* The algorithm assumes roughly spherical clusters of equal variance. A Gaussian Mixture Model (GMM) with unrestricted covariance matrices would relax this assumption and provide soft, probabilistic assignments.

(d) *Temporal snapshot.* The data reflect a single year (2024). Panel-data methods would reveal whether states are converging toward or diverging from the vassal threshold over time.

8.3 Normative Considerations

It bears emphasis that *classification is not condemnation.* The COFA nations negotiated their agreements voluntarily and derive substantial benefits—economic transfers, physical security, and access to US labour markets—from them. The analytical utility of the vassal label lies in identifying the *structural conditions* of dependency, not in passing judgement on the wisdom of the arrangements that created it.

9 Conclusion

This article has demonstrated that the concept of vassal statehood, suitably formalised in a metric feature space, admits a rigorous and empirically grounded operationalisation. The Dependency Index $\mathcal{D}$, derived from seven quantitative features and calibrated by K-Means clustering ($k = 2$), identifies a compact cluster of four COFA-bound Pacific entities—Guam, Palau, the Marshall Islands, and the Federated States of Micronesia—as the genuine vassal states of the contemporary American strategic order.

The formal proof of Theorem 4.2 establishes that this classification is optimal in the sense of minimising within-cluster dependency variance, Lemma 4.4 guarantees finite algorithmic convergence, and Corollary 4.3 quantifies the stability margin of each assignment. The PGF/TikZ visualisations confirm a clean geometric separation in feature space.

Future work should extend the dataset to all 178 countries with recorded US troop presence, incorporate time-series data to study trajectories of dependency, and apply Gaussian Mixture Models and hierarchical agglomerative clustering as complementary methods.

References

[1] Vine, D. (2015). *Base Nation: How U.S. Military Bases Abroad Harm America and the World.* Metropolitan Books.

[2] US Department of Defense, Defense Manpower Data Center (2024). *Military Personnel Statistics: Worldwide Manpower Distribution by Geographical Area.* Retrieved from https://www.dmdc.osd.mil.

[3] United Nations General Assembly (2024). *Voting Data and Records.* Retrieved from https://digitallibrary.un.org.

[4] Ganshof, F. L. (1964). *Feudalism* (3rd ed., trans. P. Grierson). Harper & Row.

[5] Wallerstein, I. (1974). *The Modern World-System, Vol. I: Capitalist Agriculture and the Origins of the European World-Economy in the Sixteenth Century.* Academic Press.

[6] Chomsky, N. (2003). *Hegemony or Survival: America's Quest for Global Dominance.* Metropolitan Books.

[7] League of Nations (1919). *Covenant of the League of Nations*, Art. 22. Paris Peace Conference.

[8] United Nations (1945). *Charter of the United Nations*, Arts. 2(1), 75–91. San Francisco.

[9] US Congress (2004). *Compact of Free Association Amendments Act of 2003*, Pub. L. 108-188, 117 Stat. 2720.

[10] Congressional Research Service (2024). *The Compacts of Free Association: A Legal and Policy Overview.* Report R44433. Library of Congress.

[11] US Agency for International Development (2024). *Foreign Aid Explorer: Annual Disbursements by Country.* Retrieved from https://explorer.usaid.gov.

[12] Lloyd, S. P. (1982). Least squares quantization in PCM. *IEEE Transactions on Information Theory*, **28**(2), 129–137.

[13] MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations. In *Proceedings of the 5th Berkeley Symposium on Mathematical Statistics and Probability*, Vol. 1, pp. 281–297.

[14] Arthur, D., & Vassilvitskii, S. (2007). k-means++: The advantages of careful seeding. In *Proceedings of the 18th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pp. 1027–1035.

[15] Keohane, R. O. (1984). *After Hegemony: Cooperation and Discord in the World Political Economy.* Princeton University Press.

[16] Mearsheimer, J. J. (2001). *The Tragedy of Great Power Politics.* W. W. Norton.

[17] Mastanduno, M. (1997). Preserving the unipolar moment: Realist theories and US grand strategy after the Cold War. *International Security*, **21**(4), 49–88.

[18] Lake, D. A. (2009). *Hierarchy in International Relations.* Cornell University Press.

[19] Bishop, C. M. (2006). *Pattern Recognition and Machine Learning.* Springer.

[20] Tan, P.-N., Steinbach, M., Karpatne, A., & Kumar, V. (2019). *Introduction to Data Mining* (2nd ed.). Pearson.

Glossary

Vassal state
A polity that formally retains sovereignty but has surrendered key prerogatives—especially defence and foreign policy—to a superior power, in exchange for protection and material benefits.

Client state
A sovereign state that, while legally independent, is heavily dependent on a patron power for security, economic support, or political legitimacy. Distinguished from vassal states by the absence of formal sovereignty cession.

Compact of Free Association (COFA)
A treaty framework between the United States and three Pacific nations (Micronesia, Marshall Islands, Palau) granting the US exclusive military access and strategic denial rights in exchange for federal transfers and a defence guarantee.

Status-of-Forces Agreement (SOFA)
A bilateral treaty defining the legal status of foreign military personnel operating in a host nation's territory, including jurisdiction over criminal offences and civil liability.

Dependency Index ($\mathcal{D}$)
A scalar measure, defined in Definition 3.1, quantifying the degree of structural dependence of a country on the United States across seven weighted dimensions.

K-Means clustering
An unsupervised machine-learning algorithm that partitions n observations into k clusters by iteratively assigning each point to its nearest centroid and recomputing centroids as cluster means.

Elbow method
A heuristic for selecting the optimal number of clusters by plotting WCSS (within-cluster sum of squares) against k and identifying the "elbow"—the point of diminishing returns in WCSS reduction.

UN voting alignment
The percentage of UN General Assembly resolutions on which a country votes in agreement with the United States, used here as a proxy for foreign-policy convergence.

World-systems theory
A sociological framework (Wallerstein, 1974) dividing the global economy into *core*, *semi-periphery*, and *periphery* states based on their structural role in the capitalist world-economy.

Hegemony
The dominance of one state over others through a combination of coercive power, economic leverage, and ideological leadership, without necessarily requiring formal colonial or vassal relationships.

Protectorate
A dependent territory that nominally retains its own government but cedes control of defence and foreign affairs to a protecting power.

PGF/TikZ
A TEX-based vector graphics language and macro package for producing publication-quality diagrams, charts, and illustrations directly within LATEX documents without external rasterisation.

Lloyd's algorithm
The standard iterative procedure for K-Means clustering, consisting of alternating assignment and centroid-update steps until convergence. See Lemma 4.4.

Standardisation (Z-score)
The transformation of a variable to zero mean and unit variance defined in equation (2), enabling fair comparison across features measured in different units.

The End